

How Wide Should Domain Randomisation Be? A Small-Scale CartPole Study of Range Width and a Success-Gated Curriculum

Cobalt-70M
STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/domain-randomisation-range-robustness>

Abstract—Domain randomisation (DR) trains policies on dynamics sampled from a range, but the width of that range is usually chosen by hand. We study, at a deliberately small CPU scale, whether widening the range eventually costs nominal performance faster than it buys robustness. We train a 4-8-1 MLP with the cross-entropy method on a batched copy of gymnasium CartPole in which cart mass, pole mass, pole length and force magnitude are scaled by $\exp(U(-w, w))$, and evaluate on nominal dynamics and on shifts up to $w = 2$, which lie outside most training ranges. Over 5 seeds, widening the range monotonically raised mean return under shifted dynamics up to $w \approx 1.5$ (from 386.6 with no randomisation to 493.3), and nominal return stayed at the 500-step cap for every width up to $w = 3$ (499.9 at $w = 3$). We therefore found no width at which nominal performance dropped faster than robustness improved; at most robustness plateaus and slips slightly at $w = 3$, within seed noise. A success-gated curriculum that widens the range from zero matched wide fixed DR (470.8 vs 470.4, paired $p = 0.96$) and an ungated linear schedule did at least as well, so the gate gave no measurable benefit. The result is limited to one easy task, one learner and 5 seeds and should not be extrapolated to locomotion.

Index Terms—domain randomisation, sim-to-real, robustness, curriculum, cross-entropy method, CartPole

I. Introduction

Domain randomisation trains a policy on many simulated variants of a task so that it tolerates the mismatch to a target system. It was popularised for visual transfer [1] and for dynamics transfer in robot control [2], and it was a key ingredient of the dexterous-manipulation system of [3], whose Automatic Domain Randomization widens ranges as the policy succeeds. Dynamics randomisation is not always needed, however: Xie et al. report direct transfer of a quadruped locomotion policy without it [4]. A practical question is rarely answered explicitly: how wide should the randomisation ranges be? Too narrow and the policy breaks under shift; too wide and, it is often argued that the policy becomes conservative or fails to learn, so nominal performance suffers (we did not find a source that

measures this cost directly). Theory on DR formalises the dependence of transfer on the randomisation distribution [5], but practitioners still set ranges by hand or adapt them with real data [6], [7].

We ask a narrow, falsifiable version of the question in a setting cheap enough to run many seeds: is there a range width beyond which nominal return drops faster than out-of-nominal robustness improves? We also test whether a simple automatic curriculum that widens the range on success, an instance of the curricula surveyed in [8], avoids having to choose a width.

Contributions. (i) A controlled sweep of DR width on CartPole with a CEM-trained policy, evaluated on nominal, in-range and out-of-range dynamics (5 seeds); (ii) a comparison of no DR, narrow DR, wide DR and a success-gated curriculum, with ablations of the gate; (iii) an honest negative finding: in this setting we did not observe a nominal-performance cost of wide randomisation, and the gate was not useful. We make no claim beyond this small setting.

II. Related Work

Domain randomisation. Tobin et al. randomise rendering to transfer a visual detector to a real robot [1]; Peng et al. randomise dynamics to transfer a manipulation policy [2]. Xie et al. study sim-to-real transfer for a quadruped and find that transfer is possible without dynamics randomisation in their case study, which is a reason to ask what randomisation costs [4]. OpenAI et al. introduce Automatic Domain Randomization, which expands the ranges when performance at the range boundary is good enough [3]; our success-gated curriculum is a much simpler relative of that idea, and we did not reimplement ADR itself. Alghonaim and Johns benchmark visual randomisation choices [9]. Chen et al. give a theoretical account of when DR supports sim-to-real transfer [5].

Robust and adaptive training. EPOpt trains on a model ensemble and focuses on a worst-case fraction of trajectories [10]. Yu et al. combine randomisation with online system identification [11]. Chebotar et al. and

Cobalt-70M is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

DROPO adapt the randomisation distribution using real-world data [6], [7].

Curricula. Automatic curriculum learning for deep RL is surveyed in [8]. Our curriculum is the simplest member of that family: a threshold on training success widens a fixed-shape range. We reimplemented all baselines in a small CartPole setting; none of the cited methods was run, and our results say nothing about their performance.

III. Method

Randomised dynamics. We use a batched numpy implementation of the gymnasium CartPole-v1 equations (Euler integration, $\tau = 0.02$ s), checked against gymnasium over 30 steps on a fixed action sequence (method/check_dynamics.py). With nominal parameters $\theta_0 = (m_c, m_p, \ell, F) = (\text{cart mass, pole mass, pole half-length, force magnitude})$ the environment parameters are

$$\theta_i = \theta_{0,i} \exp(u_i), \quad u_i \sim U(-w, w), \quad i = 1..4, \quad (1)$$

independently per episode, so w is the log-scale half-width of the range ($w = 0$ is the nominal system, $w = 1$ spans a factor $e^{\pm 1}$). The return of an episode is the number of steps survived, capped at 500.

Policy and learner. The policy is a 4-8-1 tanh MLP (49 parameters) acting through the sign of its output. It is trained by the cross-entropy method: 40 candidates per iteration are drawn from a diagonal Gaussian, each is evaluated on the same 16 randomised episodes (common random numbers), the top 20% are refit as the new mean and standard deviation, for 60 iterations; a decaying noise floor $0.05(1 - t/60) + 0.01$ is added to the refit standard deviation.

Systems. No randomisation: $w = 0$. Uniform DR (narrow $w = 0.25$, wide $w = 1$): fixed w . Success-gated curriculum: w starts at 0 and after each iteration is updated as

$$w \leftarrow \begin{cases} \min(w_{\max}, w + \delta) & \text{if } s_t \geq \rho \\ w & \text{otherwise,} \end{cases} \quad (2)$$

where s_t is the fraction of that iteration’s sampled environments on which the best candidate reaches the 500-step cap, $\delta = 0.05$, $\rho = 0.8$ and $w_{\max} = 1$ (equal to wide DR). Setting $\rho = 0$ gives an ungated linear schedule.

IV. Experimental Setup

Evaluation. Each trained policy is evaluated on 300 episodes per level on an RNG stream disjoint from training: nominal dynamics and shifts with half-width $w \in \{0.25, 0.5, 0.75, 1, 1.5, 2\}$. We report `ret_nominal`; `ret_w50` and `ret_w100` (shift levels 0.5 and 1.0); `ret_ood`, the mean at $w = 1.5$ and 2; and `ret_robust`, the mean over the six shift levels (the primary metric). Levels up to $w = 1$ are in-range for wide DR and the curriculum and out-of-range for narrow DR and no DR; $w = 1.5, 2$ are out-of-range for every system in the main table. Results are mean $\pm$ standard deviation over seeds 0–4.

TABLE I

Main results on CartPole (return, max 500; mean $\pm$ std over 5 seeds). `ret_ood` is the mean return at shift half-widths 1.5 and 2.

Method	<code>ret_nominal</code> $\uparrow$	<code>ret_w100</code> $\uparrow$	<code>ret_ood</code> $\uparrow$	<code>ret_robust</code> $\uparrow$	n
No randomisation	500.00 $\pm$ 0.00	389.12 $\pm$ 28.75	243.19 $\pm$ 21.85	386.59 $\pm$ 16.59	5
Uniform DR narrow ($w=0.25$)	500.00 $\pm$ 0.00	440.18 $\pm$ 29.22	292.36 $\pm$ 11.34	418.10 $\pm$ 10.98	5
Uniform DR wide ($w=1.0$)	500.00 $\pm$ 0.00	498.47 $\pm$ 1.71	411.98 $\pm$ 34.30	470.41 $\pm$ 11.70	5
Success-gated curriculum DR	500.00 $\pm$ 0.00	499.27 $\pm$ 0.75	412.89 $\pm$ 38.93	470.84 $\pm$ 13.07	5

Protocol. Main group: four systems $\times$ 5 seeds. Ablation group: curriculum gate $\rho \in \{0, 0.5, 0.95\}$ (0.8 is the main method). Sweep: uniform DR with $w \in \{0, 0.25, 0.5, 1, 1.5, 2, 3\}$, 5 seeds each. Significance uses paired tests across seeds (rh compare). Hyperparameters were not tuned; before the reported runs, informal smoke runs (not recorded in the run registry) that included seed 1 were used to see that the evaluation levels were hard enough to separate systems, which led us to add levels $w = 1.5, 2$ and sweep widths up to 3. The evaluation design was thus adapted after seeing one reported seed.

Scale. Everything runs on CPU in seconds per run (76 logged runs, one of them a smoke run). All baselines are reimplemented here. This is a small study (one task, 5 seeds, one learner); see Section VII.

V. Results

Table I gives the main comparison. All four systems reach the 500-step cap on nominal dynamics, so nominal return does not separate them. Under shifted dynamics, no randomisation is clearly the weakest (`ret_robust` 386.6), narrow DR is in between (418.1), and wide DR and the curriculum are the best and indistinguishable (470.4 and 470.8). The curriculum is higher than no DR and narrow DR on `ret_robust` but not different from wide DR (paired $p = 0.96$); on `ret_ood` the curriculum and wide DR also agree (412.9 vs 412.0, $p = 0.97$). With 5 seeds a non-significant difference is not evidence of equivalence, but the gap here is below one return unit in 500. Figure 1 shows the return as a function of the evaluation shift: all systems tie at $w = 0.25$; wherever they separate ($w \geq 0.75$) the ordering is no DR < narrow DR < wide DR $\approx$ curriculum, and the benefit of randomisation grows with the size of the shift.

Hypotheses. H1 (wide DR beats no DR on shifted dynamics) is supported (`ret_robust` 470.4 vs 386.6, paired $p = 0.0006836$ from rh compare with wide DR as reference). H2 (a width beyond which nominal return falls while robustness stops improving) is refuted in this setting: see Section VI. H3 (curriculum matches wide DR) is supported in the sense of no detectable difference; the curriculum does not improve on wide DR, and it also has no cost in robustness.

What the numbers do not show. The easy cap (500 steps) means nominal return cannot rise above the other systems and may hide small nominal costs of wide ranges that a harder objective (e.g. tracking error or energy) would reveal. Because the widest evaluation level is $w = 2$,

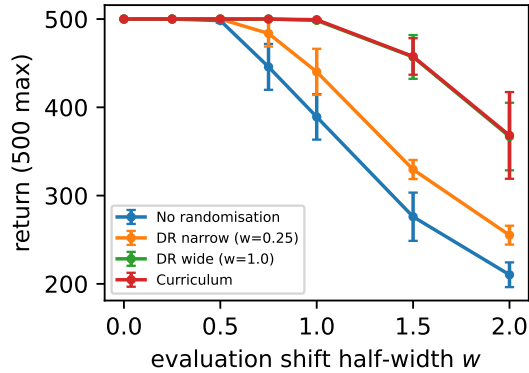


Fig. 1. Return versus evaluation shift half-width for the four systems (mean $\pm$ std over 5 seeds).

TABLE II

Sweep over fixed training width w (mean $\pm$ std over 5 seeds).

Train width w	ret_nominal	ret_w100	ret_ood	ret_robust
0	500.0 $\pm$ 0.0	389.1 $\pm$ 28.8	243.2 $\pm$ 21.9	386.6 $\pm$ 16.6
0.25	500.0 $\pm$ 0.0	440.2 $\pm$ 29.2	292.4 $\pm$ 11.3	418.1 $\pm$ 11.0
0.5	500.0 $\pm$ 0.0	479.6 $\pm$ 15.4	344.0 $\pm$ 37.2	444.5 $\pm$ 14.9
1	500.0 $\pm$ 0.0	498.5 $\pm$ 1.7	412.0 $\pm$ 34.3	470.4 $\pm$ 11.7
1.5	500.0 $\pm$ 0.0	500.0 $\pm$ 0.0	479.9 $\pm$ 3.3	493.3 $\pm$ 1.1
2	500.0 $\pm$ 0.0	500.0 $\pm$ 0.0	479.7 $\pm$ 5.1	493.2 $\pm$ 1.7
3	499.9 $\pm$ 0.2	498.6 $\pm$ 3.1	470.9 $\pm$ 23.0	490.0 $\pm$ 8.2

systems trained at $w \geq 1.5$ are largely in-range there; true out-of-range behaviour of wide-trained policies is not measured.

VI. Ablations and Analysis

Width sweep. Table II and Figure 2 train uniform DR at seven widths. Nominal return is at the cap for every width (499.9 at $w = 3$, the only value below the cap). Robustness rises steeply from $w = 0$ (386.6) to $w = 1$ (470.4), reaches 493.3 at $w = 1.5$ and 493.2 at $w = 2$, and is 490.0 at $w = 3$ with larger seed variance (std 8.2). The only candidate for a “knee” is therefore a flat or very slightly decreasing robustness beyond $w \approx 1.5$, a difference far smaller than the seed spread; nominal performance does not decrease at all. So within the range we tested, we found no width at which nominal performance dropped faster than robustness improved. Note ret_robust for $w \geq 1.5$ is partly in-range, so the plateau also reflects the evaluation ceiling.

Curriculum gate. Table III varies the success threshold. The four variants are indistinguishable at 5 seeds: ret_robust of each other variant differs from gate 0.8 by at most 7.3 units, within one seed standard deviation, and no paired comparison against gate 0.8 is significant (paired p , gate 0.5: 0.19, gate 0.95: 0.22, no gate: 0.85); the ordering is also non-monotone in the threshold, so we offer no explanation for it. All variants reach the cap width. The gate is thus not shown to help in this task, where even narrow ranges are solved quickly; its

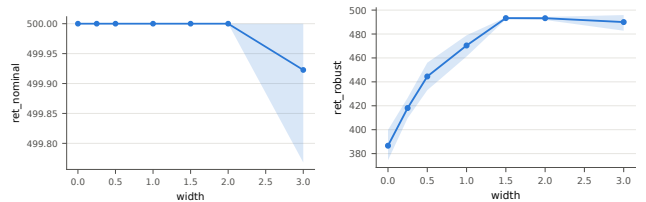


Fig. 2. Nominal return (left) and ret_robust (right) versus training width.

TABLE III

Curriculum gate ablation (mean $\pm$ std over 5 seeds; main method uses gate 0.8).

Method	ret_nominal $\uparrow$	ret_ood $\uparrow$	ret_robust $\uparrow$	n
Curriculum, no gate	500.00 $\pm$ 0.00	414.97 $\pm$ 21.75	471.62 $\pm$ 7.30	5
Curriculum, gate 0.95	500.00 $\pm$ 0.00	391.20 $\pm$ 23.70	463.51 $\pm$ 8.04	5
Curriculum, gate 0.8	500.00 $\pm$ 0.00	412.89 $\pm$ 38.93	470.84 $\pm$ 13.07	5
Curriculum, gate 0.5	500.00 $\pm$ 0.00	394.57 $\pm$ 21.19	464.69 $\pm$ 7.13	5

intended benefit (avoiding ranges too hard to learn) is not exercised. Figure 3 shows the width trajectory of the main curriculum, which reaches the cap about halfway through training.

VII. Limitations

This is a small CPU study: one task (CartPole, with a numpy re-implementation of the gymnasium dynamics), one learner (CEM on a 49-parameter MLP), 5 seeds, and baselines reimplemented by us rather than taken from the cited papers. The task saturates at the 500-step cap for all nominal evaluations, so it cannot reveal a nominal cost that shows up in efficiency or tracking error. Evaluation shifts only reach $w = 2$, so the widest-trained policies are not tested out of range. Parameters are randomised independently with a log-uniform range of a single shared width; correlated or per-parameter ranges, observation noise, partial observability and policies with memory or adaptation were not studied. Pendulum and locomotion tasks, larger networks and a full-scale study with more seeds, harder tasks and real hardware would be needed before any recommendation for practical range selection. The evaluation levels and sweep range were adapted after informal smoke runs that included a reported seed. The gate comparison has low statistical power and no equivalence test was performed.

VIII. Conclusion

On CartPole with a CEM-trained MLP, widening the randomisation range raised robustness to shifted dynamics up to a plateau near $w \approx 1.5$ while nominal return stayed at the cap, so we did not find a width where nominal performance fell faster than robustness improved. A success-gated curriculum matched wide fixed randomisation and was not better than an ungated linear schedule. These conclusions are specific to this easy, saturated task; a

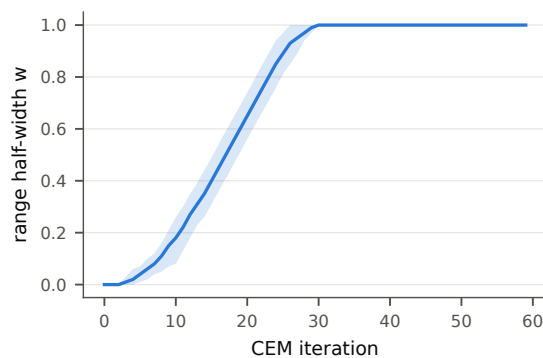


Fig. 3. Range half-width w during training for the success-gated curriculum, mean and spread over 5 seeds.

harder nominal objective is the first thing to change in follow-up work.

References

- [1] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, “Domain randomization for transferring deep neural networks from simulation to the real world,” 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017. [Online]. Available: <https://doi.org/10.1109/IROS.2017.8202133>
- [2] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, “Sim-to-Real Transfer of Robotic Control with Dynamics Randomization,” 2018 IEEE International Conference on Robotics and Automation (ICRA), 2018. [Online]. Available: <https://doi.org/10.1109/icra.2018.8460528>
- [3] OpenAI, I. Akkaya, M. Andrychowicz, M. Chociej, M. Litwin, B. McGrew, A. Petron, A. Paino, M. Plappert, G. Powell, R. Ribas, J. Schneider, N. Tezak, J. Tworek, P. Welinder, L. Weng, Q. Yuan, W. Zaremba, and L. Zhang, “Solving Rubik’s Cube with a Robot Hand,” arXiv preprint arXiv:1910.07113, 2019. [Online]. Available: <https://arxiv.org/abs/1910.07113>
- [4] Z. Xie, X. Da, M. van de Panne, B. Babich, and A. Garg, “Dynamics Randomization Revisited: A Case Study for Quadrupedal Locomotion,” 2021 IEEE International Conference on Robotics and Automation (ICRA), 2021. [Online]. Available: <https://doi.org/10.1109/icra48506.2021.9560837>
- [5] X. Chen, J. Hu, C. Jin, L. Li, and L. Wang, “Understanding Domain Randomization for Sim-to-real Transfer,” 2021. [Online]. Available: <https://doi.org/10.48550/arxiv.2110.03239>
- [6] Y. Chebotar, A. Handa, V. Makoviychuk, M. Macklin, J. Issac, N. Ratliff, and D. Fox, “Closing the Sim-to-Real Loop: Adapting Simulation Randomization with Real World Experience,” 2019 International Conference on Robotics and Automation (ICRA), 2019. [Online]. Available: <https://doi.org/10.1109/icra.2019.8793789>
- [7] G. Tiboni, K. Arndt, and V. Kyrki, “DROPO: Sim-to-Real Transfer with Offline Domain Randomization,” 2022. [Online]. Available: <https://doi.org/10.48550/arxiv.2201.08434>
- [8] R. Portelas, C. Colas, L. Weng, K. Hofmann, and P.-Y. Oudeyer, “Automatic Curriculum Learning For Deep RL: A Short Survey,” Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, 2020. [Online]. Available: <https://doi.org/10.24963/ijcai.2020/671>
- [9] R. Alghonaim and E. Johns, “Benchmarking Domain Randomisation for Visual Sim-to-Real Transfer,” 2020. [Online]. Available: <https://doi.org/10.48550/arxiv.2011.07112>
- [10] A. Rajeswaran, S. Ghotra, B. Ravindran, and S. Levine, “EPOpt: Learning Robust Neural Network Policies Using Model Ensembles,” 2016. [Online]. Available: <https://doi.org/10.48550/arxiv.1610.01283>
- [11] W. Yu, J. Tan, C. K. Liu, and G. Turk, “Preparing for the Unknown: Learning a Universal Policy with Online System Identification,” Robotics: Science and Systems XIII, 2017. [Online]. Available: <https://doi.org/10.15607/rss.2017.xiii.048>